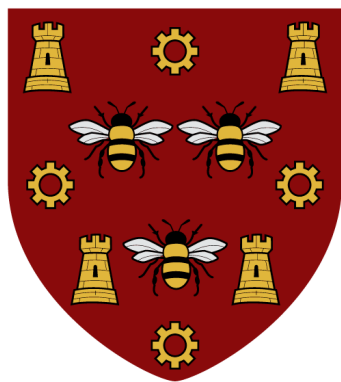




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Enhancing educational adaptability: a review and analysis of AI-driven adaptive learning platforms

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Enhancing Educational Adaptability: A Review and Analysis of AI-Driven Adaptive Learning Platforms

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Abstract- This study explores the transformative potential of AI-powered adaptive learning platforms (ALPs) in education, specifically focusing on personalized learning paths and their impact on student engagement and outcomes. Through a comprehensive analysis of four prominent ALPs—Carnegie Learning, DreamBox Learning, Smart Sparrow, and Knewton—this study investigates their approaches to content tailoring and feedback delivery. The comparative analysis highlights each platform's strengths and limitations, providing educators with valuable insights for informed selection and implementation. This study also considers the broader landscape of ALPs, acknowledging concerns such as bias, data privacy, and the role of educators in the tech-driven educational environment. The findings contribute to our understanding of how ALPs can empower educators, personalize learning, and address achievement gaps, offering a nuanced perspective on the complex tapestry of AI in education.

Keywords: Adaptive Learning Platforms (ALPs), Personalized Learning Paths, Student Engagement, Education, Intelligence.

1. INTRODUCTION

Adaptive learning is a method of education that personalizes the learning experience for each student based on their individual needs and strengths. It uses technology, such as artificial intelligence and algorithms, to assess a student's understanding and then adjust the learning materials and activities accordingly. For centuries, education has grappled with the challenge of the singular classroom. A teacher, standing before a diverse assembly of minds, strives to impart knowledge, hoping it finds fertile ground in every soil. Yet, the tapestry of learning needs and styles unfolds in vibrant and intricate patterns, often outstripping the capacity of a one-size-fits-all approach. Here, artificial intelligence-driven adaptive learning emerges, weaving a new thread into the educational fabric.

Traditionally, education followed a linear path, a river of knowledge flowing through standardized lessons and assessments. AI-driven adaptive learning, however, opens the windows of personalized learning, transforming this

river into a dynamic network of tributaries. Adaptive learning platforms (ALPs) leverage the power of AI to create individualized learning journeys, tailoring content, feedback, and pacing to the unique needs, strengths, and weaknesses of each student. These platforms act as intricate cartographers, charting the intricate landscape of each learner's mind, and guiding them along paths that maximize their engagement and understanding. Providing different levels of difficulty for different students. For example, a student who is struggling with a concept might be given more practice problems, whereas a student who is already mastering it might be given more challenging tasks. Offering different types of learning materials. Some students might learn best by reading text, whereas others might prefer watching videos or listening to audio recordings. Adaptive learning can provide students with various materials to choose from, so they can learn in the way that works best for them. Giving personalized feedback. Adaptive learning systems can track a student's progress and provide them with feedback on their strengths and weaknesses. This feedback can help students identify areas where they need to improve and focus their learning efforts. This research embarks on a comprehensive exploration of this transformative shift in the educational landscape. We delve into the intricate mechanisms of AI-driven adaptive learning by dissecting the algorithms that track student progress, identify knowledge gaps, and recommend personalized learning resources. We analyze the diverse methodologies employed by prominent ALPs and examine their approaches to content curation, real-time feedback, and dynamic course adjustments. This comparative analysis elucidates the strengths and limitations of each platform, providing educators with invaluable insights for informed selection and implementation.

However, the tapestry of ALPs is not without its knots. Concerns of bias – where algorithms may unknowingly perpetuate existing inequalities – necessitate careful consideration. Data privacy and ethical implications of student data collection require critical scrutiny. Moreover, the role of the human educator must be carefully woven into this tech-driven landscape. ALPs are not intended to replace

teachers, but rather to empower them by becoming powerful tools in their arsenals, augmenting their expertise, and enabling them to personalize learning at a scale previously unimaginable. Ultimately, this research seeks to answer a crucial question: how can AI-driven adaptive learning be harnessed to enhance the adaptability of education, bridge achievement gaps, and empower every student to reach their full potential? We believe that ALPs hold the key to unlocking a future where education is not a rigid mold, but a dynamic canvas, tailored to the unique needs of every learner. This future promises not only to revolutionize pedagogy but also to democratize access to knowledge, ensuring that every student, regardless of background or circumstance, has the opportunity to embark on a personalized and engaging learning journey.

The main contributions of this paper are

- This study investigated the effectiveness of AI-powered learning platforms, focusing on student engagement and learning outcomes.
- This study undertakes a comprehensive comparison of a few renowned ALPs and dissects their mechanisms for better educational adaptability.
- The findings suggest that adaptive learning can improve student learning outcomes compared with traditional teaching methods in terms of knowledge retention and skill development.

II. RELATED WORKS

Gligorea et al. (2023) explored the integration of AI/ML into e-learning platforms for adaptive learning and assessed its impact and challenges. The study found that AI/ML improved engagement, retention, and learning outcomes, but highlighted challenges such as data quality, ethical considerations, and cost implementation.[1] Jing et al. (2023) analyzed research trends in adaptive learning between 2000 and 2022, identifying key actors, publications, and research themes to understand the field's development. Four key research themes emerged: applying deep learning in educational data analysis, developing new adaptive learning models, integrating intelligent tutoring systems, and advancing feature modeling techniques.[2] Wang et al. (2020) compared the effectiveness of an adaptive learning system (Squirrel AI Learning) with traditional teacher-led instruction (large-group and small-group) in Chinese eighth-grade math classes. The study found that students who used Squirrel AI Learning achieved significantly higher math test scores than those who participated in traditional teacher-led classes, regardless of class size.[3] Dabingaya (2022) investigated the effectiveness of AI-powered adaptive learning platforms in mathematics education, focusing on student engagement and learning outcomes. The AI group exhibited increased engagement and achieved significantly higher post-assessment scores, suggesting that the platform improved learning outcomes and supported personalized learning pathways.[4] Yu et al. (2017) explored how emerging AI techniques could have personalized learning in Massive Open Online Courses (MOOCs) to improve the learning experience and expand research opportunities. They argued that AI-powered personalization could enhance MOOCs by tailoring learning paths, maximizing teaching assistant efficiency, and offering richer student interaction, while also yielding

valuable data for educational research.[5] Bloom et al. (2021) investigated the feasibility and effectiveness of implementing adaptive learning within higher education institutions using the Moodle LMS platform. The case study demonstrated positive student engagement with the adaptive course, indicating the potential of Moodle LMS as a viable tool for personalized learning in higher education settings.[6] Costa et al. (2021) examined the effectiveness of personalized and adaptive learning approaches in educational settings by comparing them with traditional teaching methods. This paper also highlighted the challenges and suggested future research directions for optimizing the technology's potential.[7] Kem (2022) reviewed emerging personalized and adaptive learning platforms in the digital and smart learning era, assessing their capabilities and potential benefits. The review concluded that personalized and adaptive learning platforms offered promising benefits for individual learning pace, skill mastery, and engagement, suggesting a bright future for their integration into diverse educational settings.[8] Kabudi et al. (2021) understood the landscape of AI-powered adaptive learning systems (ALPs) by analyzing publications from 2014 to 2020. The analysis revealed a diverse application of AI in ALPs, identified common challenges such as data privacy, and highlighted the potential for personalized learning and improved student outcomes.[9] Yaghmaie and Bahreininejad (2011) proposed a new framework for an adaptive learning management system (LMS) using multi-agent technology and semantic web ontologies. The research showed promise for the proposed approach in dynamically adapting learning content to individual needs, exceeding the limitations of traditional personalization methods within LMS.[10]

III. PROPOSED MODEL OF AN ADAPTIVE LEARNING PLATFORM(ALP)

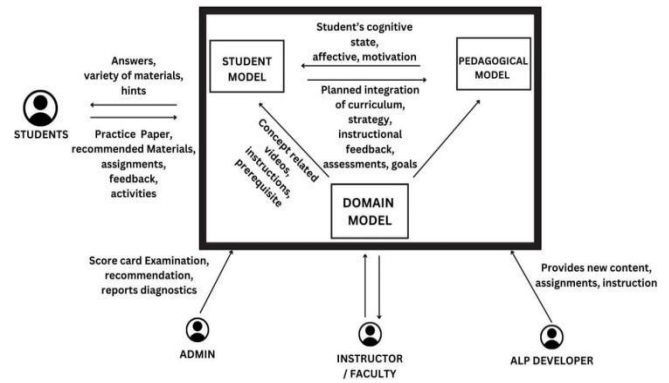


Fig. 1. Proposed model and operation of an adaptive learning platform

Adaptive learning platforms work by tailoring the learning experience to each individual user's needs and level of understanding. Fig 1. shows the working of an adaptive learning platform, which may vary according to different platforms and their goals. Platforms often start with an initial assessment to gauge a learner's prior knowledge, skills, and learning style. This can involve quizzes, surveys, or interactive activities. As the learner progresses, the platform continuously collects data on their performance through assessments, interactions, and time spent on different topics. This data is analyzed by algorithms, often powered by artificial intelligence, to identify the learner's strengths,

weaknesses, and preferred learning methods. Based on the analysis, the platform dynamically adapts the learning path, tailoring it to the individual's needs. This might involve: Adjusting the difficulty level of the content. Providing targeted practice in areas needing improvement. Offering alternative learning materials based on the preferred styles. Recommending additional challenges for advanced learners. The platform provides immediate feedback on the learners' progress, helping them understand their strengths and areas for improvement. This can motivate and guide them further. As the learner continues to interact with the platform, the process of data collection, analysis, and adaptation repeats itself, ensuring a constantly evolving and personalized learning experience. Faculty still play a crucial role in the learning process. They are subject matter experts who can create high-quality, engaging content aligned with curriculum objectives. They can tailor content to specific student needs and learning styles beyond what algorithms can capture. They can connect platform learning to real-world applications and challenges, making it more meaningful.

IV. COMPARATIVE ANALYSIS OF ALPs

With the rise of personalized learning in the digital age, educators increasingly seek adaptive learning platforms (ALPs) to effectively cater to individual needs and enhance educational adaptability. This study delves into a comparative analysis of four prominent ALPs: Carnegie Learning, DreamBox Learning, Smart Sparrow, and Knewton, examining their approaches to tailoring content and delivering feedback.

1) Carnegie Learning



Fig 2. User Interface of Carnegie Learning

Carnegie Learning empowers personalized learning pathways within a structured curriculum as shown in Fig 2. Their platform uses "knowledge spaces" to map student understanding and dynamically adjust content difficulty. By analyzing interactive exercises and quizzes, the system recommends tailored learning activities to fill knowledge gaps and promote mastery. This focus on individual progress leads to improved engagement, knowledge retention, and overall achievement. While pre-built content provides structure, adaptive elements ensure that each student progresses at their own pace, creating a customized learning journey toward mastery.

2) DreamBox Learning



Fig 3. User Interface of DreamBox Learning

DreamBox Learning personalizes the K-12 math learning experience using adaptive technology as shown in Fig 3. Imagine a tutor constantly adjusting to your understanding, filling your knowledge gaps, and challenging you just right. That is DreamBox in action. It analyzes your responses to interactive exercises, identifying strengths and weaknesses. These data fuels a custom learning path that offers targeted practice problems, hints, and explanations tailored to your needs. Like a dream guide, it leads you toward mastery, adapting to your pace and learning style to keep you engaged and motivated.

3) Smart Sparrow

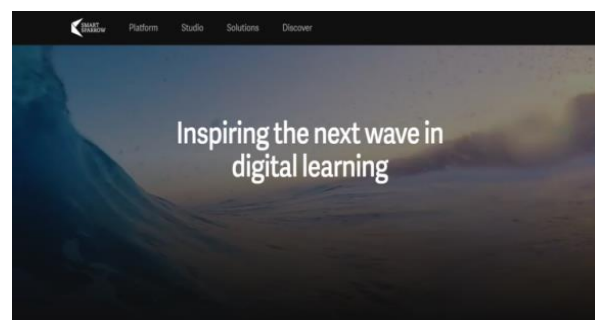


Fig 4. User Interface of Smart Sparrow

As shown in Fig 4. Smart Sparrow empowers educators to create interactive and personalized learning experiences that foster deeper engagement and knowledge retention. Imagine building custom activities with simulations, branching scenarios, and multimedia content. As students interact, Smart Sparrow's AI engine assesses their understanding using various techniques such as Bayesian knowledge tracing and reinforcement learning. Based on this, it dynamically adapts the content, offering personalized feedback, hints, or branching paths to optimize the learning journey. It's like having a smart tutor tailored to each student's needs, guiding them toward mastery at their own pace.

4) Knewton

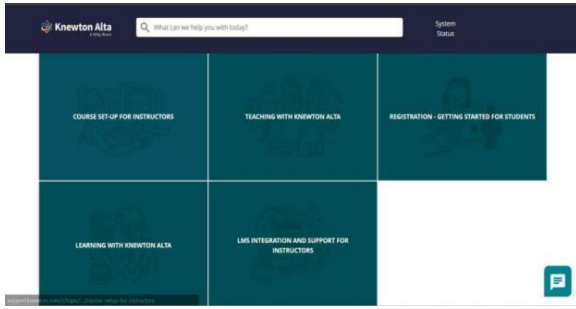


Fig 5. User Interface of Knewton

As shown in Fig 5. Knewton Learning uses an adaptive learning platform powered by a sophisticated algorithm. Imagine a student entering a virtual forest of knowledge. At each turn, the platform assesses students' understanding through interactive questions [11-15]. Based on the responses, the algorithm selects the most suitable learning path, offering personalized content and challenges. It adapts in real-time, filling knowledge gaps and reinforcing strengths. Think of it as a dynamic tutor, constantly evaluating and guiding students on their unique learning journey, aiming for optimal mastery and avoiding redundant practice. This personalized approach empowers students to progress at their own pace, maximizing their learning potential.

Through a thorough comparative analysis, as shown in Table 1. below, we delve into their respective strengths and limitations in fostering educational adaptability, providing educators with valuable insights for informed platform selection and implementation.

TABLE I. Comparative analysis of different ALPS

ALPS	Principle	Pros	Cons
Carnegie Learning	Real-time data analysis informs teaching and curriculum adjustments. Students progress at their own pace, mastering concepts before moving on.	Students progress at their own pace, mastering concepts before moving on. Adaptive algorithms help close the gap between high- and low-performing students.	Carnegie Learning primarily focuses on mathematics and science, with limited offerings in other subjects. Carnegie Learning is more expensive than traditional textbooks or other educational resources.

DreamBox Learning	Personalized learning paths are created on the basis of student performance and cognitive data. Gamification elements such as rewards and challenges motivate students and make learning fun.	Gamification elements make learning math enjoyable and reduce anxiety. The platform encourages critical thinking and a deep understanding of math concepts.	DreamBox can be expensive for some schools and families, thus limiting access. Currently focuses primarily on K-8 math, excluding other subjects and higher education levels.
Smart Sparrow	The platform emphasizes rich multimedia content and interactive elements to engage learners and cater to diverse learning styles. Content is delivered in bite-sized modules that focus on active learning and spaced repetition for better knowledge retention.	The interactive and visually rich content fosters active learning and keeps students engaged throughout the learning process. Smart Sparrow offers features such as text-to-speech and keyboard navigation, making it accessible to learners with disabilities.	Some features may require specific hardware or software, potentially creating barriers for some users. Smart Sparrow can be more expensive than traditional learning platforms, making it less accessible to certain institutions.
Knewton	Knewton leverages AI algorithms to personalize learning paths based on individual student strengths and weaknesses. Students progress through content based on demonstrating competence and not simply completing assignments.	Each student receives a tailored learning journey that caters to their specific needs and pace. Educators gain valuable insights into student performance and areas requiring additional support.	Knewton's licensing can be expensive compared with traditional educational resources. Knewton primarily focuses on STEM subjects, leaving other fields with limited options.

V. EVALUATING ALPS BY COMPARING DIFFERENT PARAMETERS

Table 2. shows the comparison between different ALPS on the basis of different parameters such as their ratings, response time, and privacy. From the given table, we can conclude that the employee rating of Carnegie Learning is better than that of other given adaptive learning platforms. Carnegie Learning also surpasses other ALPs in terms of the privacy of data given to their customers; personal information is unsold or rented to the third parties. The

response time of Carnegie Learning also outshines that of other ALPs.

Table 2. Comparing ALPS on the basis of their different specifications

ALPS	Ratings	Response Time	Privacy
Carnegie Learning	3.9	253 ms	91.4%
DreamBox Learning	2.5	987 ms	88.3%
Smart Sparrow	3.8	551 ms	80.8%
Knewton	3.5	709 ms	52.1%

The performance of different ALPS are evaluated with different ensemble based predictive methods like decision tree, random forest, boosting and bagging. It was observed that modeling with boosting method recorded the best outcome in context to the mean f-score metric with a value of 91.7%. Table 2 shows the summarized outcome.

Table 3. F-Score metric analysis of different ALPs

ALPS	F-score (Boosting)
Carnegie Learning	92.4%
DreamBox Learning	90.8%
Smart Sparrow	90.2%
Knewton	93.7%

VI. CONCLUSION

AI-powered adaptive learning platforms (ALPs) represent a dynamic element within the fabric of contemporary education. This study meticulously examined the intricate workings of these platforms, revealing their capacity to individualize learning paths, provide tailored feedback, and adapt dynamically to diverse student needs. The observed impact on educational adaptability is noteworthy because it offers more effective and engaging learning experiences for students with varied styles and abilities. However, this intricate educational tapestry is not immune to challenges. The considerations of bias, data privacy, and the evolving role of educators in a technology-driven landscape require careful attention. To unlock the full potential of ALPs, ongoing research and development are imperative to ensure responsible application and continual improvement.

In the future collaboration among educators, researchers, and technology developers is paramount. The focus should be on harnessing AI’s potential to empower educators, narrow achievement gaps, and foster a genuinely inclusive learning environment. The success of ALPs, hinges not on replacing teachers but on enhancing their expertise, weaving a dynamic tapestry of personalized learning experiences for every student.

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